INCH-POUND

MIL-D-51410(EA)
AMENDMENT 1
22 February 1989

MILITARY SPECIFICATION

DYE, GREEN, ETHYL-BIS(2,4-DINITROPHENYL) ACETATE

This amendment forms a part of MIL-D-51410(EA), dated 22 October 1974, and is approved for use by the U.S. Army Chemical Research, Development and Engineering Center, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

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3.1: Delete Table 1 in its entirety and substitute the following:

"Table 1. Chemical and physical characteristics

Characteristics	Requirements		Test paragraph
	Minimum	Maximum	
Absorbance, percent of standard	97.5	----	4.2.4.1
Melting point, °C	147	156	4.2.4.2.
Solids content, percent by weight	50	----	4.2.4.3

"

AMSC N/A

FSC 6820

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3.2: Add the following: "The dye used in making the paper shall pass through a 44 micron (No. 325) sieve with not more than 15 percent by weight retained on the sieve."

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4.2.2.2: Delete 4.2.2.2 and table IV in their entirety and substitute the following:

"**4.2.2.2 For test.** Each drum shall be tested for solids content. Sampling for all other tests shall be conducted by making a composite from the dried samples obtained from the solids content tests. The composite sample shall be tested in duplicate."

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4.2.4.1:

Line 1: Delete "**Molar absorbtivity**" and substitute "**absorbance**" and add the following as the first sentence in the paragraph: "Determine the absorbance of the standard sample (see 6.7) and test the sample as follows:"

Lines 13 through 15: Delete this calculation in its entirety and substitute the following:

"
$$\text{Percent absorbance} = \frac{\text{Absorbance of sample}}{\text{Absorbance of standard}} \times \frac{\text{Weight standard}}{\text{Weight sample}}$$
"

4.2.4.3: Delete in its entirety and substitute the following:

"**4.2.4.3 Solids content.** Weigh to the nearest milligram approximately 10g of specimen into a tared open dish. Dry to constant weight at 115° ± 5°C. Cool and weigh specimen and dish to the nearest milligram. Calculate the percent solids as follows:

$$\text{Percent solids} = \frac{B-C}{A-C}(100)$$

where: A = weight of specimen and dish before heating, in grams
B = weight of specimen and dish after heating, in grams
C = weight of dish, in grams "

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4.2.4.4: Delete in its entirety.

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5.1: *Delete first sentence and substitute the following:*

"A quantity of 200 (+5) pounds of dye shall be packed in a 3 mil polyethylene liner in a nominal 30 gallon capacity 24 gauge steel drum furnished with a gasketed, fully removable cover. One tail drum containing less than 200 pounds is permissible, per lot."

5.2 (e): Add the following; "percent solids, pounds dye".

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Section 6: Add the following:

"**6.7 Standard sample.** A standard dye sample for use in the absorbance test may be obtained from Commander, U.S. Army Armament, Munitions and Chemical Command, Product Assurance Directorate, Chemical Systems Division, Detection Branch, Aberdeen Proving Ground, MD 21010-5423, refer to 'Dye Green EDA Standard 45161.' "

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